Early-Stage Financial Modeling of Industrial Investment Projects

v2.0.0 — Johannes Aurich — johannesaurich.com

September 7, 2026

About this document: *I've been involved in techno-economic analyses of industrial equipment projects; this is a documentation of the key concepts and mathematical modeling approaches I used and derived to simplify the early-stage comparative analysis of project configurations. Rigorous mathematical notation is not applied. This document is meant as a personal reminder of key concepts, without the ambition to be easy-to-understand for anyone else. I nevertheless hope that it can be helpful to others.*

Contents

1 Basic Concepts

1.1 Nominal vs. Real Financial Value

Simply said, nominal financial values are those printed on receipts and bank notes, stated on accounts, and usually used to calculate taxes. They represent the exchange value of goods and services in a specific year. Real values, on the other hand, reflect the actual purchasing power of money. They consider that inflation erodes monetary value over time and express how much a nominal value in a future year t is worth relative to a base year $t = 0$. Assuming a constant inflation rate π over a time period t, the real value $V_{r,t}$ of a nominal financial value $V_{n,t}$ in year t is defined as:

$$V_{r,t} := \frac{V_{n,t}}{(1+\pi)^t} \tag{1}$$

As can be seen in equation 1, nominal and real financial values are identical in the base year and inflation effects are zero. Real values and inflation rates are always defined relative to the base year, which is often chosen as the present day in most calculations but can generally be any other reference year depending on the context. Real-world inflation rates typically vary over time and may require more complex modeling approaches than discussed here.

1.2 Present Value

The present value PV is a key financial metric for investment projects and usually applied to cash flows. It can be defined in real or nominal terms, however nominal values are more typical because of easier accessibility and straightforward use. It adjusts, or discounts, a cash flow C_t in year t by a discount rate d relative to a base year $t = 0$:

$$PV := \frac{C_t}{(1+d)^t} \tag{2}$$

Note: Depending on the chosen approach, both cash flow and discount rate must be defined in either nominal or real terms to ensure consistency. Discounting e.g. nominal cash flows with real rates is nonsensical.

Equation 2 shows that this is conceptually related to the conversion between real and nominal values. However, beyond adjusting for inflation, the discount rate usually includes additional premiums for project risk, time value of money, and financing costs. This means the present value is often lower than the real monetary value, providing a more realistic value assessment of future financial transactions. The discount rate is often calculated using the concept of WACC (weighted average cost of capital), which won't be further discussed here. Adapted to the specific context, the present value concept is widely applied across industries.
If instead of a single cash flow a series of cash flows over multiple years must be considered, the overall present value is obtained by summing the present value of all individual cash flows:

$$PV = \sum_{t=0}^{T} \frac{C_t}{(1+d)^t} = \sum_{t=0}^{T} PV_t \tag{3}$$

This additive property is very useful if we consider that every cash flow of a year may itself have multiple components. For example, the concept of the **Net Present Value (NPV)** defines each cash flow as the difference of all revenues R_t and all expenses E_t in that year: $C_t := R_t - E_t$. We can then determine the NPV by calculating and then summing the present values of revenues and expenses separately:

$$NPV := \sum_{t=0}^{T} \frac{R_t - E_t}{(1+d)^t} \tag{4}$$

$$NPV = \sum_{t=0}^{T} \frac{R_t}{(1+d)^t} - \sum_{t=0}^{T} \frac{E_t}{(1+d)^t} \tag{5}$$

$$= \sum_{t=0}^{T} PV(R_t) - \sum_{t=0}^{T} PV(E_t) \tag{6}$$

This allows to single out and treat individual value contributors separately and, depending on the problem, simplify the overall approach. For example, a machine may require the replacement of key components as an overhauling investment in years $t = (5, 8)$, additional to the annual operating expenses. Multiple service providers and spare part vendors shall be compared. A simple approach is to calculate the present value of all candidate offers (equation 3: $PV = PV_5 + PV_8$) and then choose the most economic one. No other cashflows must be considered.

Note: The present value is not restricted to cash flows. One can equally calculate a present value of material, energy, or other quantities that are relevant to judge the performance of a process or machine, using the exact same formulas. A typical application is the levelized costs of energy (LCOE) concept.

2 Deriving Simplified Annuity Factors

Converting between present values and annually recurring cash flows is a common problem in financial analysis. Generally, you must evaluate equation 3 for each year separately but early-stage financial modeling often allows you to use simplified assumptions. There are two scenarios where specific annuity factors, convert between annually recurring cash flows and present values, can be derived: if the annual cash flows are either **(a) constant** or **(b) changing exponentially**.

The *Cumulative Discount Factor* CDF calculates the present value PV of a recurring and constant annual cash flow $\bar{C}$, while the *Capital Recovery Factor* CRF, as its inverse, determines which constant annual cash flow is equivalent to a given present value:

$$CDF := \frac{PV}{\bar{C}} \qquad CRF := \frac{\bar{C}}{PV} \qquad CRF = CDF^{-1} \tag{7}$$

Both CDF and CRF will be derived in the following sections, covering both the constant and exponentially changing annual cash flow scenarios.

Note: It is crucial to keep in mind that both CRF and CDF involve discounting calcuations and whether nominal or real annuities are used is highly important. This will be addressed in section 2.3.

2.1 Scenario (a): Constant Periodic Cash Flows

Starting with the mathematical definition of the present value in equation 3, we can make simplifying assumptions that will be further explained and justified in section 3:

- There is no cash flow C_t before year $t = \tau$: $\rightarrow C_t = 0 \quad \forall t \in \{0, 1, \ldots, (\tau - 1)\}$, common if installation or commissioning is required before equipment can operate.
- All cash flows C_t after start to the end of operation at $t = T$ are constant: $\rightarrow C_t = \bar{C} = \mathrm{const} \quad \forall t \in \{\tau, (\tau + 1), \ldots, T\}$.
- All annual discount rates d are constant in all years.

This allows us to rewrite equation 3 and separate the sum at $t = \tau$ to remove all summands $C_t = 0$. Factoring out the constant cash flows $C_t = \bar{C}$ and introducing a *Discount Factor* k leads to:

$$PV = \sum_{t=0}^{T} \frac{C_t}{(1+d)^t} = \underbrace{\sum_{t=0}^{\tau-1} \frac{C_t}{(1+d)^t}}_{=0} + \sum_{t=\tau}^{T} \frac{C_t}{(1+d)^t} \tag{8}$$

$$= \sum_{t=\tau}^{T} \frac{C_t}{(1+d)^t} = \bar{C} \sum_{t=\tau}^{T} \frac{1}{(1+d)^t} = \bar{C} \sum_{t=\tau}^{T} k^t \tag{9}$$

$$k = \frac{1}{(1+d)} \tag{10}$$

An explicit solution for the sum $\sum_{t=\tau}^{T} k^t$ can be found if one recognizes the similarity to the finite geometric series:

$$\sum_{n=0}^{N} a^n = \frac{1 - a^{N+1}}{1 - a} \quad \text{for} \quad a \neq 1 \tag{11}$$

The finite geometric series includes terms from $t = 0 \ldots T$, while the sum in this scenario only ranges from $t = \tau \ldots T$. This can be solved by subtracting the excluded first part of the sum from $t = 0 \ldots (\tau - 1)$:

$$\sum_{t=\tau}^{T} k^t = \sum_{t=0}^{T} k^t - \sum_{t=0}^{\tau-1} k^t = \frac{1 - k^{T+1}}{1 - k} - \frac{1 - k^{\tau}}{1 - k} \tag{12}$$

$$= \frac{k^{\tau} - k^{T+1}}{1 - k} \quad \text{for} \quad k \neq 1 \tag{13}$$

The definitions of CDF and CRF from equation 7 can now be specified for $k \neq 1$:

$$CDF = \frac{PV}{\bar{C}} = \sum_{t=1}^{T} k^t = \frac{k^\tau - k^{T+1}}{1-k} \tag{14}$$

$$CRF = \frac{\bar{C}}{PV} = \left(\sum_{t=1}^{T} k^t\right)^{-1} = \frac{1-k}{k^\tau - k^{T+1}} \tag{15}$$

The form of equation 15 differs from other more broadly used formulations of the CRF; section 2.4 provides a proof for their equivalence under specific circumstances.

2.2 Scenario (b): Exponential Escalation of Periodic Cash Flows

It has been assumed above that all periodically recurring cash flows, and in a generalized sense also non-monetary recurring values, are constant in all years. However, an additional factor can model degradation of machinery outputs and related revenues or any regular price escalation. Under the assumption that this change occurs with a constant annual escalation rate r, each annual value becomes:

$$C_t = \bar{C}(1+r)^t \tag{16}$$

Mind that degradation or decrease in general means $r < 0$ and increase is $r > 0$. We can now modify equation 9 and derive a new discount factor:

$$PV = \sum_{t=\tau}^{T} \frac{\bar{C}(1+r)^t}{(1+d)^t} = \bar{C} \sum_{t=\tau}^{T} \left(\frac{1+r}{1+d}\right)^t \tag{17}$$

$$\rightarrow k = \left(\frac{1+r}{1+d}\right) \tag{18}$$

Using the same approach, an arbitrary number N of escalation or degradation rates r_i can be applied to a recurring value $\bar{C}$:

$$k = \frac{(1+r_1)(1+r_2)(1+r_3)\ldots}{(1+d)} = \frac{1}{(1+d)} \prod_{i=1}^{N} (1+r_i) \tag{19}$$

2.3 Differentiating between Real and Nominal Approaches

So far, no distinction was made between nominal or real vaules because the derived equations are valid for both. In practice, the CRF is more commonly used with *nominal* values while the CDF is more useful with *real* values. The reasons are simple:

- CRF_n, *nominal recovery*: When an investment is made or debt taken on at $t = 0$ as known present value, the recovery through annuities is usually fixed in contracts or business plans and executed in concrete financial transactions. This favors the use of nominal annuities to explicitly define the exact amount due every year.

- CDF_r, *real discounting*: Estimating the present value of future expenses or revenues requires assumptions about the time value of money. Often, the only thing you can reasonably assume is that market participants will try to maintain their purchasing power, meaning keeping the real value of transactions constant by adjusting their nominal values annually (further assuming the intrinsic real value of goods and services do not change). This allows you to forecast based on real revenues and expenses from base year $t = 0$ onwards; which are, by definition, equal to the nominal values at $t = 0$ and thereby know.

This can lead to mistakes because $CRF = 1/CDF$ is only true if both use nominal $()_n$ or real rates $()_r$. Mind that this includes discount rates d as well as any escalation rates r_i, etc. The correct relations are:

$$CRF_n \neq CDF_r^{-1} \quad | \quad d_n \neq d_r \quad r_{i,n} \neq r_{i,r} \tag{20}$$
$$CRF_n = CDF_n^{-1} \quad | \quad d = d_n \quad r_i = r_{i,n} \tag{21}$$
$$CRF_r = CDF_r^{-1} \quad | \quad d = d_r \quad r_i = r_{i,r} \tag{22}$$

The Fisher equation allows to convert between nominal and real rates α by relating them to the annual inflation rate π, $\alpha \in \{d, r_i\}$:

$$(1 + \alpha_n) = (1 + \pi)(1 + \alpha_r) \tag{23}$$

2.4 Limitations of Widely Used Equations

At first glance, equation 15 looks different than usual CRF definitions in textbooks or online which typically use the discount rate d instead of the discount factor k:

$$CRF = \frac{d(1+d)^T}{(1+d)^T - 1} \tag{24}$$

As is shown below, both expressions are identical for scenario (a), meaning constant cash flows without any exponential escalation fators ($r_i = 0$) and $k = 1/(1+d)$ as in equation 10. Rearranging to $d = (1/k) - 1$ and substituting in equation 24 leads to the generalized CRF definition in equation 15 for $\tau = 1$.

$$CRF = \frac{\left(\frac{1}{k} - 1\right)\left(1 + \frac{1}{k} - 1\right)^T}{\left(1 + \frac{1}{k} - 1\right)^T - 1} = \frac{\left(\frac{1}{k} - 1\right)\left(\frac{1}{k}\right)^T}{\left(\frac{1}{k}\right)^T - 1} = \frac{\left(\frac{1}{k}\right)^T}{\left(\frac{1}{k}\right)^T} \cdot \frac{\left(\frac{1}{k} - 1\right)}{1 - \frac{1}{\left(\frac{1}{k}\right)^T}} \tag{25}$$

$$= \frac{\left(\frac{1-k}{k}\right)}{1 - k^T} = \frac{1-k}{k - k^T} = CRF(\tau = 1)\big|_{\text{Eq.15}} \qquad \square \tag{26}$$

This equality does not hold for scenario (b) with $r_i \neq 0$ and all $\tau \neq 1$. This means that the popular definition of the CRF in equation 24 is only a special case of the more general and versatile definition in equation 15.

3 Modeling Early-Stage Industrial Investments

Financial models for large-scale industrial investments can be arbitrarily complex and detailed. Early-stage models are mostly concerned with techno-economic trade-off studies to identify the right project configuration and determine their general financial viability. Each product and project is different, so the assumptions made during modeling must be adapted to every case individually.
The scenario presented below is a pragmatic approach for many types of industrial equipment, from production machinery to grid-scale batteries, and can be extended and modified easily to match the specifics of each case. The key assumptions were already hinted in section 2.1 and will be justified more extensively in the following.
To determine the financial performance of a project configuration, it is useful to think about the present value of a project PV_{Pr} as sum of the present values of all investments PV_{In} and all operating cash flows PV_{op} during the project lifetime. The assumptions listed below are based on this approach.

$$PV_{pr} = PV_{in} + PV_{op} \tag{27}$$

Note: The general assumption is that all annual rates d and r_i are constant in all years.

3.1 Assumption (1): Delayed Commercial Operation

There is usually a significant gap between purchase and start of commercial operation because complex machinery requires extensive installation and commissioning work, software must be included into existing IT systems and users trained before roll out, and some projects come with additional construction effort. This means a project can be split into two basic phases:

1. *Pre-operation*, where investments are made and the purchased assets are delivered and commissioned before the commercial operation date (COD): $0 \leq t_{in} \leq t_{COD}$.

2. *Commercial operation*, every year after reaching the commercial operation date (COD) until the equipment's end of life (EOL): $t_{COD} \leq t_{op} \leq t_{EOL}$.

3.2 Assumption (2): Predicable Real Operating Cash Flows

The nominal value of goods and services will change over time due to inflation and changes in market structure, but in which way is hard to foresee. A pragmatic assumption for the commercial operation phase in early-stage models is to assume that the market environment and thereby the intrinsic value of goods and services will stay constant or change predictably, meaning all other changes in nominal prices are caused by inflation only (market participants will adjust prices for inflation to preserve their purchasing power). Two scenarios aligned with sections 2.1 and 2.2 can be considered:

(a) The intrinsic value of goods and services doesn't change and all prices on a real basis are constant in all years: $C_{t,op} = \bar{C}_{r,op} = const.$

(b) There is a predictable annual percentage change in intrinsic values (e.g. from foreseeable worsening supply shortages, asset degradation, etc.) which can be modeled through escalation factors: $C_{t,op} = \bar{C}_{r,op} \prod_{i=1}^{N} (1+r_i)^t$ with $\bar{C}_{r,op} = const.$

If neither of these two scenarios apply to the whole range of $C_{t,op}$, it is often possible to split $C_{t,op}$ into multiple components and time periods which fit above scenarios, like different revenues and expense streams happening at different times with individual escalation behavior. Changing annual rates can be handled similarly.
The benefit of this approach is that, usually, the value of machine inputs and outputs in the modeling base year t=0 (energy, utilities, raw materials, labor to products and services) can be determined from quotes and first principle calculations; and because nominal prices are equal to real prices in the base year by definition, these known values can be used as modeling basis: $\bar{C}_{r,op} \equiv C_{0,op}$.

Using above assumptions, we can calculate the present value of operational cash flows using the CDF_r (equation 14) with $\tau = t_{COD}$ and $T = t_{EOL}$ as well as the real discount factor k_r (equation: 19), with $r_{r,i} = 0$ for scenario (a) without and $r_{r,i} \neq 0$ for scenario (b) with escalation of the real values:

$$PV_{op} = \sum_{t=t_{COD}}^{t_{EOL}} \frac{C_{t,op}}{(1+d_r)^t} = C_{0,op} \cdot CDF_r \tag{28}$$

$$\text{with} \quad CDF_r = \frac{k_r^{t_{COD}} - k_r^{t_{EOL}+1}}{1-k_r} \tag{29}$$

$$\text{Scenario (a):} \quad k_r = \frac{1}{(1+d_r)} \tag{30}$$

$$\text{Scenario (b):} \quad k_r = \frac{1}{(1+d_r)} \prod_{i=1}^{N} (1+r_{r,i}) \tag{31}$$

3.3 Assumption (3): Milestone or Lump-Sums Investments

Investments $C_{in,t}$ are non-recurrent payments and can be made in two ways:

1. Pre-operation as initial CAPEX investments to purchase the equipment.
2. During commercial operation for refurbishment or replacement of faulty parts.

Those payments are either transfered as single lump-sums or split into multiple installments based on the achievement of milestones, e.g. the timely delivery of key components or the successful completion of commissioning tests by the supplier. But because they're non-recurring, their present value can only be calculated by discounting every payment individually over the whole project duration $0 \leq t \leq t_{EOL}$, using equation 3:

$$PV_{in} = \sum_{t=0}^{t_{EOL}} \frac{C_{in,t}}{(1+d)^t} \tag{32}$$

For the special case that a single initial lump-sum payment at $t = 0$ makes up all the invested capital, the equation simplifies to $PV_{in} = C_{in,0}$.

3.4 Key Metrics: NPV and IRR

Project economics are often estimated based on the *Net Present Value* NPV and *Internal Rate of Return* IRR. The concept of the NPV has been mentioned before, it is defined as present value sum of all project revenues R_t and project expenses E_t:

$$NPV := \sum_{t=0}^{T} \frac{R_t - E_t}{(1+d)^t} \tag{33}$$

$$= \sum_{t=0}^{T} PV(R_t) - \sum_{t=0}^{T} PV(E_t) \tag{34}$$

Using the assumption from sections 3.2 and 3.3, revenues are operating incomes while expenses are either operating expenses (OPEX), like raw materials and energy purchases, or investments/capital expenditurs (CAPEX) throughout the whole project lifetime. The sale of equipment for its residual value at t_{EOL} may be treated as a non-recurring revenue.

$$NPV = PV_{op,revenues} - PV_{op,expenses} - PV_{in} \tag{35}$$

The IRR, on the other hand, is a metric based on the NPV. It is the discount rate d for which the NPV becomes exactly zero:

$$NPV = 0 = \sum_{t=0}^{T} \frac{R_t - E_t}{(1+IRR)^t} \tag{36}$$

This equation must usually be solved iteratively.